

Intro to Fractals and Complex Dynamics

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Abstract

In this talk, we look at various kinds of fractals, how they're formed and their properties. We also explore important objects in complex dynamics, and finish up with results on a particular rational map $z^n + \frac{a}{z^d} + c$.

1 What are Fractals?

Before talking about complex dynamics, we first need to discuss fractals.

There are many kinds of fractals all generated differently, but there are several properties that occur frequently.

1. Self-similarity: smaller pieces resemble the whole object.
2. Structure at arbitrarily small scales: one can zoom in to find new details.
3. Non-integer dimension: under certain definitions of "dimension," a fractal may have dimension between the usual integer dimensions.
4. Simple recursive constructions: complicated shapes can come from repeating a simple rule many times.

The final point will be especially important in this talk. Many fractals in complex dynamics arise simply by repeatedly applying the same function.

1.1 Some Examples

Serpinski Triangle:

1. Begin with an equilateral triangle.
2. Divide it into four congruent smaller triangles.
3. Remove the center triangle.
4. Repeat the process on each remaining triangle.

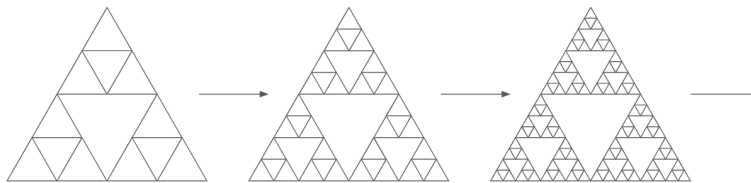


Figure 1: Sierpinski Triangle Construction

The Sierpiński triangle also gives an example of a non-integer dimension. For a line segment, scaling by $\frac{1}{2}$ gives two smaller copies: $2 = 2^1$, and for a square, scaling by $\frac{1}{2}$ gives four smaller copies: $4 = 2^2$. For the Sierpiński triangle, scaling by $\frac{1}{2}$ produces three copies. If its dimension is (D), we obtain $3 = 2^D$, so $D = \frac{\log 3}{\log 2} \approx 1.585$. In this sense, the Sierpiński triangle has dimension between that of a line and that of a plane. It's also clearly self-similar and has arbitrarily small detail, and is generated by repeating a simple set of rules.

Fractals can also be not self-similar. Take any real-world object outline, for example. Ignoring the fact that there's some limit to how far we can "zoom", because anything physical is inherently rough in some way, one can zoom in further and further and find a new texture. This object can be created by the simple rule of adding more and more sides to some polygon that further resembles this object. And lastly, it has a slightly more convoluted, albeit proper, definition for dimension. These more complicated fractals will be important later in the talk.

2 Dynamic Systems

2.1 Real Inputs

A dynamical system describes how an object changes over time. For our purposes, we consider discrete dynamical systems, in which we repeatedly apply the same function. Suppose

$$f(x) = x^2.$$

Starting at $x = 2$, we obtain $2 \mapsto 4 \mapsto 16 \mapsto 256 \mapsto \dots$, which tends quickly to infinity.

Starting instead at $x = \frac{1}{2}$, we obtain $\frac{1}{2} \mapsto \frac{1}{4} \mapsto \frac{1}{16} \mapsto \dots$, which tends to 0. These orbits can be visualized in Figure 2.

Note two things in the figure:

1. There are two fixed points, $x = 0$ and $x = 1$. A fixed point is a point satisfying $f(x) = x$. In this case, $x^2 = x$, so $x = 0$ and $x = 1$. We can determine the behavior near a fixed point using the derivative. Since $f'(x) = 2x$, we have $|f'(0)| = 0 < 1$, so 0 is an attracting fixed point,

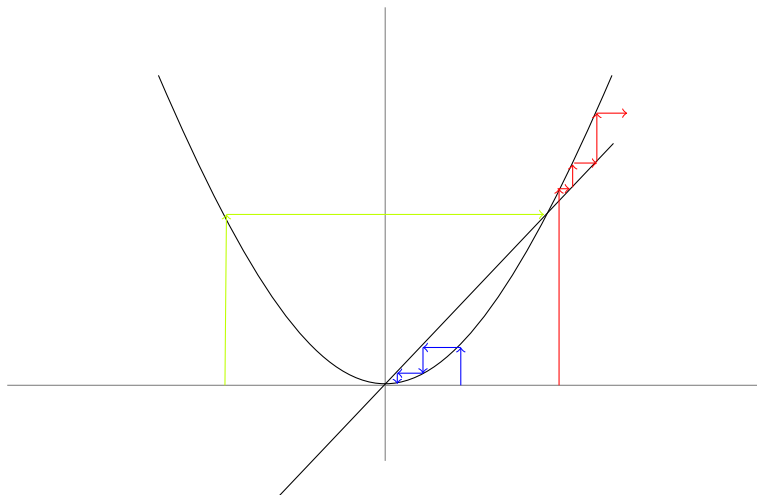


Figure 2: Orbits of $x = -1$, 0.5 , and 1.1 for $f(x) = x^2$

meaning nearby orbits are pulled toward it. On the other hand, $|f'(1)| = 2 > 1$, so 1 is a repelling fixed point, meaning nearby orbits are pushed away from it.

2. It is also possible to have periodically fixed points. A point has period k if $f^k(x) = x$. For example, if $f(x) = x^2 - 1$, then $0 \mapsto -1 \mapsto 0$. Thus, 0 and -1 form a cycle of period 2 . As with fixed points, derivatives can be used to determine whether a periodic cycle is attracting or repelling.

2.2 Complex Inputs

So far, we have only considered functions with real-valued inputs. However, we can ask the same questions when our inputs are complex numbers.

Recall that a complex number has the form $z = x + iy$, and can be represented as the point (x, y) in the complex plane.

Thus, rather than choosing a starting value on a number line, we can now choose a starting point anywhere in a plane and repeatedly apply a function to it.

Consider the family

$$f_c(z) = z^2 + c,$$

where $c \in \mathbb{C}$ is fixed. Just as in the real case, for each starting point z , we can consider its orbit $z \mapsto f_c(z) \mapsto f_c^2(z) \mapsto f_c^3(z) \mapsto \dots$. Some of these orbits remain bounded, while others escape to infinity. For example, if $c = 0$, then $f(z) = z^2$. If $|z| < 1$, then $|f(z)| = |z|^2 < |z|$, so repeated iteration pushes the orbit toward 0 . If $|z| > 1$, then $|f(z)| = |z|^2 > |z|$, so the orbit grows and

escapes to infinity. The unit circle therefore forms a boundary between these two types of behavior.

This gives us a first example of a Julia set.

Definition 2.1. For a polynomial or rational map f , the filled Julia set is

$$K(f) = \{z \in \mathbb{C} : \{f^n(z)\}_{n=0}^{\infty} \text{ is bounded}\}.$$

In other words, the filled Julia set contains all starting points whose orbits never escape to infinity.

Definition 2.2. The Julia set of f is the boundary of the filled Julia set:

$$J(f) = \partial K(f).$$

For the map

$$f(z) = z^2,$$

the filled Julia set is the closed unit disk and the Julia set is the unit circle. However, this example is unusually simple. Changing the constant c can produce dramatically more complicated Julia sets.

For example, now consider

$$f(z) = z^2 - 1.$$

We already saw in the real case that $0 \mapsto -1 \mapsto 0$ is a periodic orbit. In the complex plane, however, there are infinitely many starting points to consider, and the boundary between bounded and escaping orbits becomes much more complicated.

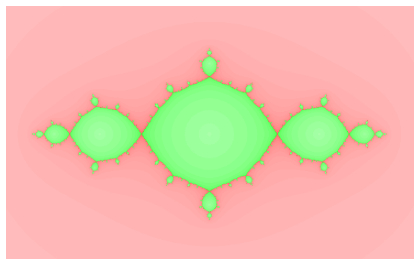


Figure 3: The filled Julia set and Julia set for $f(z) = z^2 - 1$. This Julia set is often called the Basilica.

Julia sets can look very different depending on the value of c .

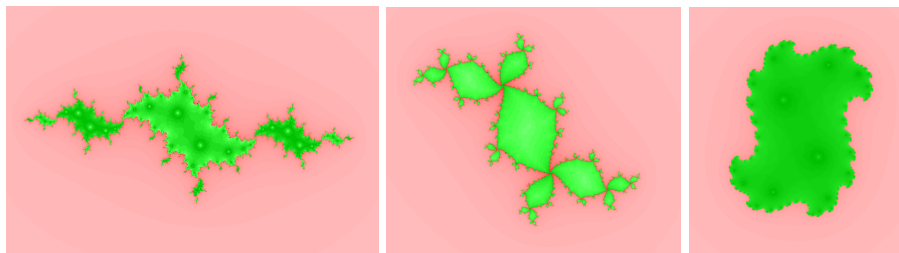


Figure 4: Several Julia sets from the family $f_c(z) = z^2 + c$ for $c = -1.1 + 0.2i$, $-0.1 + 0.75i$, $0.3 - 0.2i$, respectively.

Another question one can ask is: Instead of fixing c and varying the starting point z , what happens if we vary c itself?

2.3 Dynamical Space and Parameter Space

There are two ways to study a family of functions like $f_c(z) = z^2 + c$

1. In the dynamical plane, we fix the parameter c and vary the starting point z . Each point in the plane represents one possible initial condition for the same dynamical system. (Julia sets can live here)
2. In parameter space, we vary c . Each point in the plane now represents an entirely different function, which can itself have Julia sets in the dynamical plane.

The most famous example of a parameter space is the Mandelbrot set, the set of parameters $c \in \mathbb{C}$ for which the orbit of 0 under $f_c(z) = z^2 + c$ remains bounded. If the orbit $0 \mapsto c \mapsto c^2 + c \mapsto (c^2 + c)^2 + c \mapsto \dots$ remains bounded, then c lies in the Mandelbrot set. If it escapes to infinity, then c lies outside the Mandelbrot set.

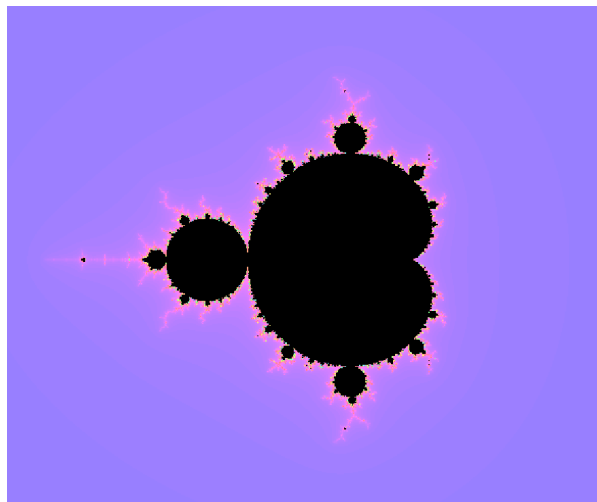


Figure 5: The Mandelbrot set. Each point c represents a different map $f_c(z) = z^2 + c$.

Note that Mandelbrot-like sets (and corresponding Julia sets) can also be generated for $f_c(z) = z^n + c$.

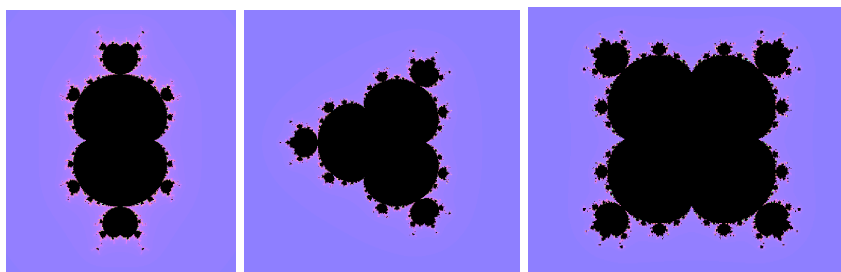


Figure 6: Mandelbrot-like sets for $f_c(z) = z^n + c$ where $n = 3, 4, 5$

We can now broaden our class of functions from polynomials to rational maps.

3 Rational Maps

So far, we have focused primarily on polynomial maps such as $f_c(z) = z^2 + c$. A rational map is a function of the form $R(z) = \frac{P(z)}{Q(z)}$ where P and Q are polynomials and Q is not zero. Every polynomial is also a rational map, since we can simply take $Q(z) = 1$.

Rational maps introduce new behavior that does not occur for polynomials. In particular, if $Q(z) = 0$, then the function has a pole. Near a pole, the

magnitude of the function can become arbitrarily large. For example, $R(z) = z^2 + \frac{1}{z^2}$ has a pole at $z = 0$.

If z is very large, then the z^2 term dominates. However, if z is very close to 0, then $\frac{1}{z^2}$ dominates. Thus, unlike the polynomial examples from earlier, points can be sent toward infinity both because they are very far from the origin and because they are very close to the origin (0 maps to infinity, in this case).

This interaction between behavior near 0 and behavior near infinity produces many interesting Julia sets.

3.1 McMullen Maps

For the following section, here's a key for the rational maps:

1. $F_{n,c}(z) = z^n + c$ (Unicritical Maps)
2. $F_{n,a}(z) = z^n + \frac{a}{z^n}$ (McMullen Maps)
3. $F_{n,c,a}(z) = z^n + \frac{a}{z^n} + c$ (Generalized McMullen Maps)
4. $F_{n,d,c,a}(z) = z^n + \frac{a}{z^d} + c$ (Maximally Generalized McMullen Maps)
5. $F_{n,d,c,a}(z) = z^n + \frac{a}{z^d} + c$, $n, d \in \mathbb{R}$ (Extremely Maximally Generalized McMullen Maps)
6. $F_{\eta,\delta,c,a}(z) = z^n + \frac{a}{z^d} + c$, $\eta, \delta \in \mathbb{R}$ (Complex Extremely Maximally Generalized McMullen Maps)

An important family of rational maps is the McMullen family, $M_{n,d,a}(z) = z^n + \frac{a}{z^d}$, where n, d are positive integers and $a \neq 0$.

3.2 The Maximally Generalized McMullen Family

The main family considered in our work is the Maximally Generalized McMullen family,

$$F_{n,d,c,a}(z) = z^n + \frac{a}{z^d} + c$$

where $n, d \geq 3$, $a \in \mathbb{C}^*$, $c \in \mathbb{C}$.

This family generalizes several of the examples we have already discussed. If $a = 0$, we recover a polynomial of the form $z^n + c$. If $c = 0$, we obtain a classical McMullen-type map with $n \neq d$: $z^n + \frac{a}{z^d}$.

For much of this talk, we will be especially interested in what happens when n and d become large.

3.3 Dynamical and Parameter Space

Just as with $f_c(z) = z^2 + c$, we can study the maximally generalized McMullen family in both the dynamical plane and parameter space.

In the dynamical plane, we fix n, d, a, c and vary the starting point z . This gives the filled Julia set $K(F_{n,d,c,a})$ and Julia set $J(F_{n,d,c,a})$.

In parameter space, we fix n, d, a and vary c . Each point in the c -plane then represents a different rational map.

Definition 3.1. For fixed n, d , and a , the boundedness locus is

$$M_{n,d,a} = \{c \in \mathbb{C} : \text{at least one critical orbit of } F_{n,d,c,a} \text{ is bounded}\}.$$

This plays a role similar to the Mandelbrot set. Recall that for $f_c(z) = z^2 + c$, the critical point is $z = 0$, and the Mandelbrot set records the parameters for which its orbit remains bounded.

For the maximally generalized McMullen family, there are many critical points, so the parameter space is more complicated.

Critical Points and Values

Critical points occur where $F'(z) = 0$. For $F(z) = z^n + \frac{a}{z^d} + c$, we have $F'(z) = nz^{n-1} - \frac{da}{z^{d+1}}$. Setting this equal to 0 gives

$$z^{n+d} = \frac{da}{n}.$$

Thus there are $n+d$ finite nonzero critical points, evenly spaced around a circle.

The images of these points under F are called the critical values. Although there are $n+d$ critical points, there are only $\frac{n+d}{\gcd(n,d)}$ distinct critical values. This comes from rotational symmetry. Let $\ell = \gcd(n,d)$ and $\omega = e^{2\pi i/\ell}$. Then $F(\omega z) = F(z)$, since $\omega^n = \omega^d = 1$. Thus, the arithmetic relationship between n and d makes a clearly visible symmetry between the Julia sets and parameter spaces. (Note that for purposes of this talk, we just refer to the parameter space of c , but using a is also possible, just more complicated.)

What Happens as n and d Grow?

One of the main questions in our work is what happens when the exponents become very large.

If $|z| > 1$, then $|z|^n$ becomes very large as n increases, so the z^n term tends to force points outside the unit circle toward infinity.

If $|z| < 1$, then $|z|^d$ becomes very small as d increases, so $|\frac{a}{z^d}|$ becomes very large. Thus points inside the unit circle are also pushed toward infinity.

These observations can be made rigorous. Roughly, for any $\varepsilon > 0$, sufficiently large n keeps the Julia set from lying far outside $|z| = 1 + \varepsilon$, while sufficiently large d keeps it from lying far inside $|z| = 1 - \varepsilon$.

Thus, for large n and d , the Julia sets are squeezed into a thin annulus around the unit circle: $1 - \varepsilon < |z| < 1 + \varepsilon$.

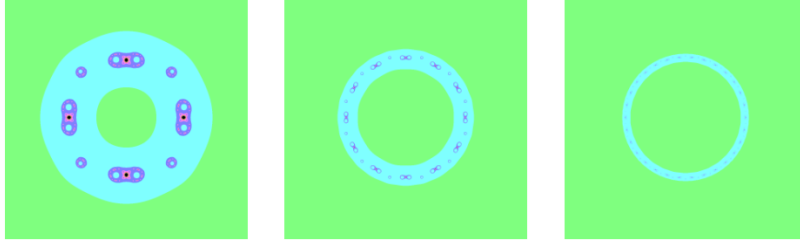


Figure 7: As n and d grow, the Julia sets are squeezed toward the unit circle.

If we also have $\gcd(n, d) \rightarrow \infty$, then the rotational symmetry increases, and the geometric limit becomes the entire unit circle.

So, very roughly, $J(F_{n,d,c,a}) \rightarrow S^1$.

Parameter Space

The critical values determine a collection of circles in the c -plane, called the spine. As n and d grow, the boundedness locus becomes increasingly concentrated around these circles.

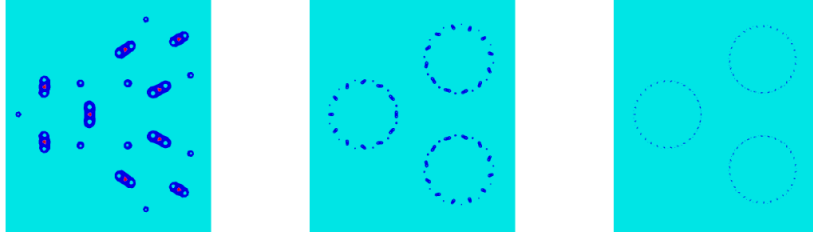


Figure 8: A boundedness locus together with its spine, with the same values as the Julia set above.

TL;DR: Julia sets $\longrightarrow$ unit circle in the dynamical plane, while boundedness loci $\longrightarrow$ collections of circles in parameter space.

In the special case $n = d$, the parameter-space limit approaches two unit circles centered at $\pm 2\sqrt{a}$.

Beyond Integer Exponents

The usual family assumes n and d are integers, but many of these ideas can also be extended to positive real exponents. One can consider

$$F_{\eta,\delta,c,a}(z) = z^\eta + \frac{a}{z^\delta} + c,$$

with $\eta, \delta \in \mathbb{R}_{>0}$.

For complex z , non-integer powers are defined using a branch of the complex logarithm, for example $z^\eta = e^{\eta \log z}$. This makes the symmetry more subtle, but many of the same broad ideas—escape near 0 and infinity, annular bounds, and circle-like structures—still have analogues.

Meanwhile, a and c may already take complex values in the integer-exponent family, so these Julia sets and parameter spaces naturally live in the complex plane.

4 References

References

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